



1. Description

1.1. Project

Project Name	ROTARY_MIXER
Board Name	custom
Generated with:	STM32CubeMX 6.10.0
Date	06/04/2025

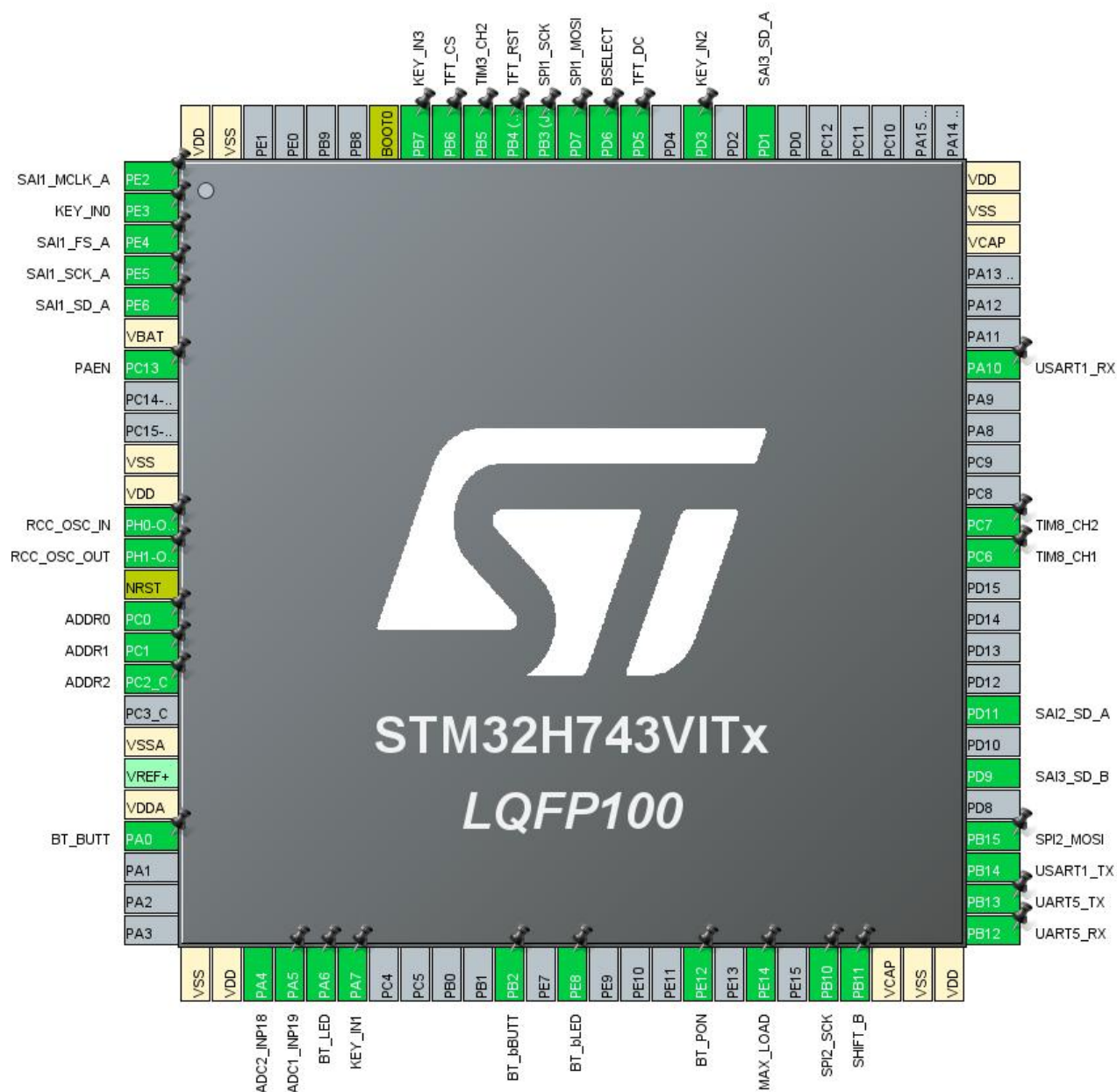
1.2. MCU

MCU Series	STM32H7
MCU Line	STM32H743/753
MCU name	STM32H743VITx
MCU Package	LQFP100
MCU Pin number	100

1.3. Core(s) information

Core(s)	ARM Cortex-M7
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2. Pinout Configuration



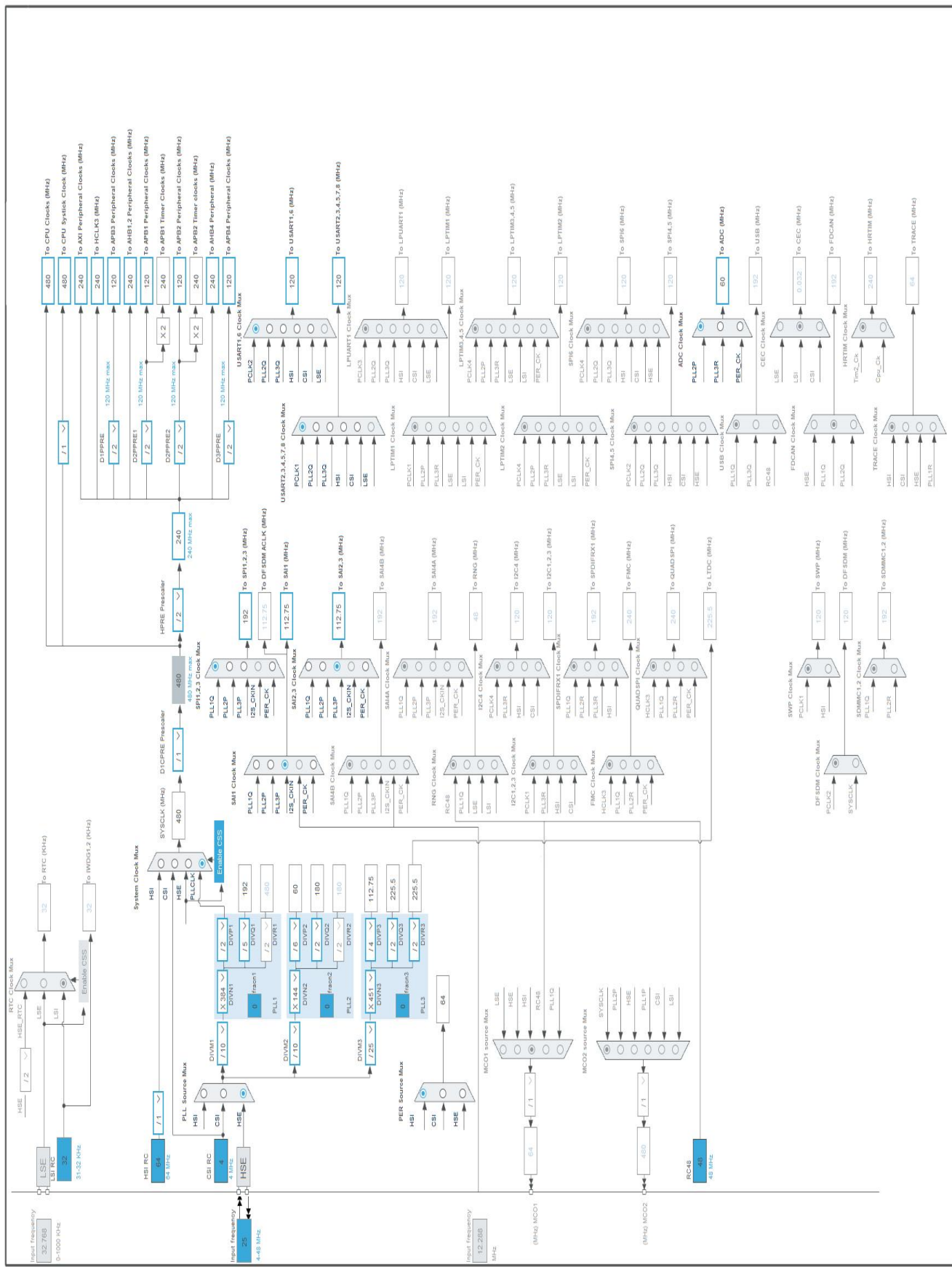
3. Pins Configuration

Pin Number LQFP100	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	PE2	I/O	SAI1_MCLK_A	
2	PE3 *	I/O	GPIO_Input	KEY_IN0
3	PE4	I/O	SAI1_FS_A	
4	PE5	I/O	SAI1_SCK_A	
5	PE6	I/O	SAI1_SD_A	
6	VBAT	Power		
7	PC13 *	I/O	GPIO_Output	PAEN
10	VSS	Power		
11	VDD	Power		
12	PH0-OSC_IN (PH0)	I/O	RCC_OSC_IN	
13	PH1-OSC_OUT (PH1)	I/O	RCC_OSC_OUT	
14	NRST	Reset		
15	PC0 *	I/O	GPIO_Output	ADDR0
16	PC1 *	I/O	GPIO_Output	ADDR1
17	PC2_C *	I/O	GPIO_Output	ADDR2
19	VSSA	Power		
21	VDDA	Power		
22	PA0 *	I/O	GPIO_Output	BT_BUTT
26	VSS	Power		
27	VDD	Power		
28	PA4	I/O	ADC2_INP18	
29	PA5	I/O	ADC1_INP19	
30	PA6 *	I/O	GPIO_Input	BT_LED
31	PA7 *	I/O	GPIO_Input	KEY_IN1
36	PB2 *	I/O	GPIO_Input	BT_bBUTT
38	PE8 *	I/O	GPIO_Output	BT_bLED
42	PE12 *	I/O	GPIO_Output	BT_PON
44	PE14 *	I/O	GPIO_Output	MAX_LOAD
46	PB10	I/O	SPI2_SCK	
47	PB11 *	I/O	GPIO_Input	SHIFT_B
48	VCAP	Power		
49	VSS	Power		
50	VDD	Power		
51	PB12	I/O	UART5_RX	
52	PB13	I/O	UART5_TX	
53	PB14	I/O	USART1_TX	

Pin Number LQFP100	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
54	PB15	I/O	SPI2_MOSI	
56	PD9	I/O	SAI3_SD_B	
58	PD11	I/O	SAI2_SD_A	
63	PC6	I/O	TIM8_CH1	
64	PC7	I/O	TIM8_CH2	
69	PA10	I/O	USART1_RX	
73	VCAP	Power		
74	VSS	Power		
75	VDD	Power		
82	PD1	I/O	SAI3_SD_A	
84	PD3 *	I/O	GPIO_Input	KEY_IN2
86	PD5 *	I/O	GPIO_Output	TFT_DC
87	PD6 *	I/O	GPIO_Output	BSELECT
88	PD7	I/O	SPI1_MOSI	
89	PB3 (JTDO/TRACESWO)	I/O	SPI1_SCK	
90	PB4 (NJTRST) *	I/O	GPIO_Output	TFT_RST
91	PB5	I/O	TIM3_CH2	
92	PB6 *	I/O	GPIO_Output	TFT_CS
93	PB7 *	I/O	GPIO_Input	KEY_IN3
94	BOOT0	Boot		
99	VSS	Power		
100	VDD	Power		

* The pin is affected with an I/O function

4. Clock Tree Configuration



5. Software Project

5.1. Project Settings

Name	Value
Project Name	ROTARY_MIXER
Project Folder	C:\Keil_v5\My_Project\XDJX_MIXER_ASSY
Toolchain / IDE	MDK-ARM V5.32
Firmware Package Name and Version	STM32Cube FW_H7 V1.11.1
Application Structure	Basic
Generate Under Root	No
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

5.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy all used libraries into the project folder
Generate peripheral initialization as a pair of '.c/.h' files	Yes
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	No
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

5.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_UART5_Init	UART5
4	MX_SAI1_Init	SAI1
5	MX_SAI2_Init	SAI2
6	MX_SAI3_Init	SAI3
7	MX_TIM2_Init	TIM2
8	MX_ADC1_Init	ADC1
9	MX_ADC2_Init	ADC2
10	MX_SPI1_Init	SPI1
11	MX_SPI2_Init	SPI2

Rank	Function Name	Peripheral Instance Name
12	MX_TIM8_Init	TIM8
13	MX_USART1_UART_Init	USART1
14	MX_TIM3_Init	TIM3

1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32H7
Line	STM32H743/753
MCU	STM32H743VITx
Datasheet	DS12110 Rev8

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Alkaline(9V)
Capacity	625.0 mAh
Self Discharge	0.3 %/month
Nominal Voltage	9.0 V
Max Cont Current	200.0 mA
Max Pulse Current	0.0 mA
Cells in series	1
Cells in parallel	1

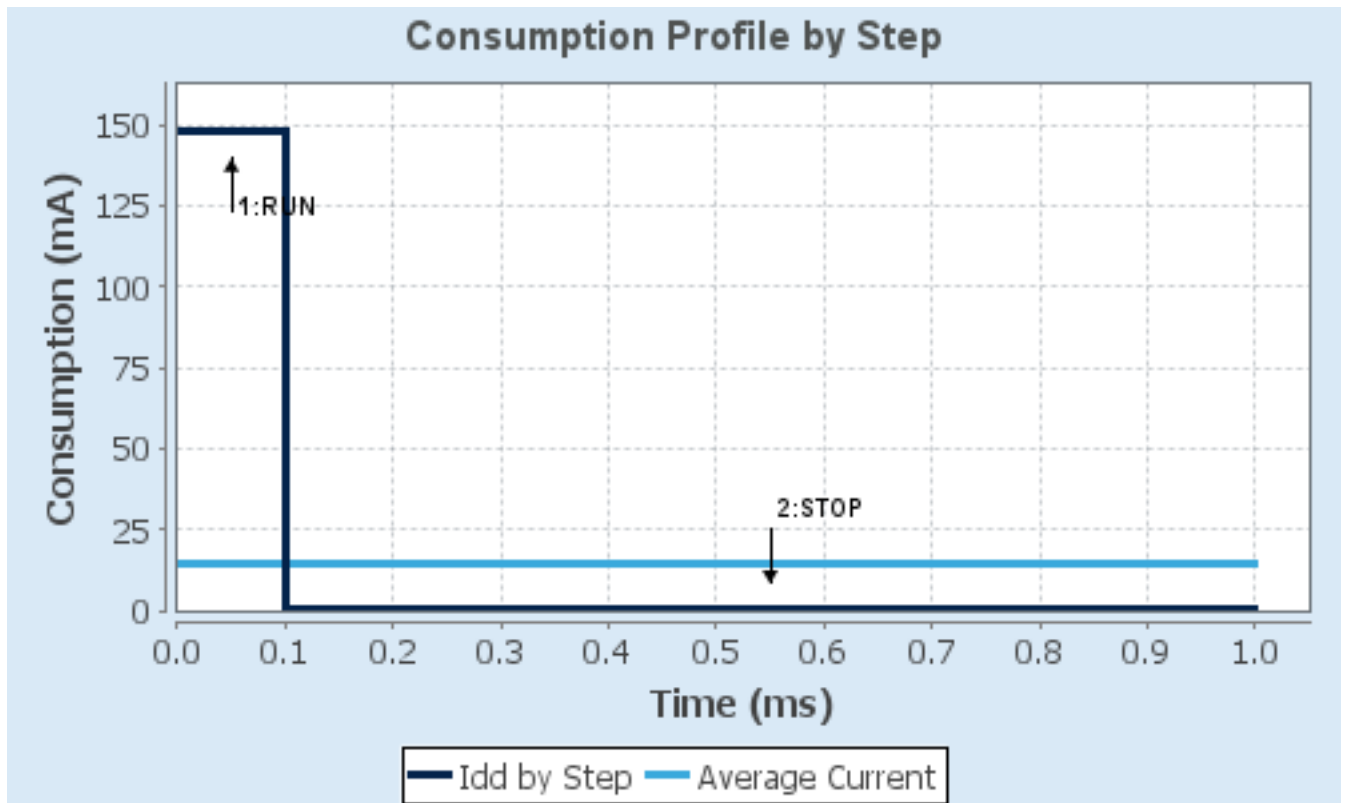
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	VOS0: Scale0-High	SVOS5: System-Scale5
D1 Mode	DRUN/CRUN	DSTANDBY
D2 Mode	DRUN	DSTANDBY
D3 Mode	DRUN	DSTOP
Fetch Type	ITCM	NA
CPU Frequency	480 MHz	0 Hz
Clock Configuration	HSE BYP PLL	Flash-OFF
Clock Source Frequency	24 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	148 mA	150 μ A
Duration	0.1 ms	0.9 ms
DMIPS	1027.0	0.0
Ta Max	105.02	124.98
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	14.94 mA
Battery Life	1 day, 17 hours	Average DMIPS	1027.2001 DMIPS

1.6. Chart



2. Peripherals and Middlewares Configuration

2.1. ADC1

mode: IN19

2.1.1. Parameter Settings:

ADCs_Common_Settings:

Mode Independent mode

ADC_Settings:

Clock Prescaler Asynchronous clock mode divided by 1

Resolution

ADC 12-bit resolution *

Scan Conversion Mode

Disabled

Continuous Conversion Mode

Disabled

Discontinuous Conversion Mode

Disabled

End Of Conversion Selection

End of single conversion

Overrun behaviour

Overrun data preserved

Left Bit Shift

No bit shift

Conversion Data Management Mode

Regular Conversion data stored in DR register only

Low Power Auto Wait

Disabled

ADC_Regular_ConversionMode:

Enable Regular Conversions

Enable

Enable Regular Oversampling

Disable

Number Of Conversion

1

External Trigger Conversion Source

Regular Conversion launched by software

External Trigger Conversion Edge

None

Rank

1

Channel

Channel 19

Sampling Time

1.5 Cycles

Offset Number

No offset

ADC_Injected_ConversionMode:

Enable Injected Conversions

Disable

Analog Watchdog 1:

Enable Analog WatchDog1 Mode

false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode

false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode

false

2.2. ADC2

IN18: IN18 Single-ended

2.2.1. Parameter Settings:

ADCs_Common_Settings:

Mode Independent mode

ADC_Settings:

Clock Prescaler Asynchronous clock mode divided by 1

Resolution

ADC 12-bit resolution *

Scan Conversion Mode Disabled

Continuous Conversion Mode Disabled

Discontinuous Conversion Mode Disabled

End Of Conversion Selection End of single conversion

Overrun behaviour Overrun data preserved

Left Bit Shift No bit shift

Conversion Data Management Mode Regular Conversion data stored in DR register only

Low Power Auto Wait Disabled

ADC_Regular_ConversionMode:

Enable Regular Conversions Enable

Enable Regular Oversampling Disable

Number Of Conversion 1

External Trigger Conversion Source Regular Conversion launched by software

External Trigger Conversion Edge None

Rank 1

Channel Channel 18

Sampling Time 1.5 Cycles

Offset Number No offset

ADC_Injected_ConversionMode:

Enable Injected Conversions Disable

Analog Watchdog 1:

Enable Analog WatchDog1 Mode false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode false

2.3. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

2.3.1. Parameter Settings:

Power Parameters:

SupplySource	PWR_LDO_SUPPLY
Power Regulator Voltage Scale	Power Regulator Voltage Scale 0

RCC Parameters:

TIM Prescaler Selection	Disabled
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000
CSI Calibration Value	32
HSI Calibration Value	64

System Parameters:

VDD voltage (V)	3.3
Flash Latency(WS)	4 WS (5 CPU cycle)
Product revision	rev.V

PLL range Parameters:

PLL1 clock Input range	Between 2 and 4 MHz
PLL2 input frequency range	Between 2 and 4 MHz
PLL3 input frequency range	Between 1 and 2 MHz
PLL1 clock Output range	Wide VCO range
PLL2 clock Output range	Wide VCO range
PLL3 clock Output range	Wide VCO range

2.4. SAI1

Mode: Master with Master Clock Out

mode: External Synchro Out

2.4.1. Parameter Settings:

SAI A:

Synchronization Inputs	Asynchronous
Protocol	Free
Audio Mode	Master Transmit
Frame Length	64 bits *
Data Size	16 Bits *
Slot Size	DataSize
Output Mode	Stereo
Companding Mode	No companding mode
SAI SD Line Output Mode	Driven
First Bit	MSB First

Frame Synchro Active Level Length	32 *
Frame Synchro Definition	Channel Identification *
Frame Synchro Polarity	Active Low
Frame Synchro Offset	Before First Bit *
First Bit Offset	0
Number of Slots (only Even Values)	2
Slot Active Final Value	0x00000003 *
Slot Active	User Setting *
Slot 0 Active	true *
Slot 1 Active	true *
Clock Source	SAI PLL Clock
Master Clock No Divider	Enabled
Audio Frequency	44.1 KHz *
Real Audio Frequency	44.042 KHz *
Error between Selected	0.09 % *
Clock Strobing	Rising Edge *
Fifo Threshold	Empty
Output Drive	Enabled *
Advanced Parameters:	
Synchronization External	Output SAI A enabled *

2.5. SAI2

Mode: Synchronous Slave

2.5.1. Parameter Settings:

SAI A:

Synchronization Inputs	Synchronous with other block of same SAI
Protocol	Free
Audio Mode	Slave Receive
Frame Length (only Even Values)	32 *
Data Size	16 Bits *
Slot Size	DataSet
Output Mode	Stereo
Companding Mode	No companding mode
SAI SD Line Output Mode	Driven
First Bit	MSB First
Frame Synchro Active Level Length	1

Frame Synchro Definition	Start Frame
Frame Synchro Polarity	Active Low
Frame Synchro Offset	First Bit
First Bit Offset	0
Number of Slots	2 *
Slot Active Final Value	0x00000003 *
Slot Active	User Setting *
Slot 0 Active	true *
Slot 1 Active	true *
Clock Strobing	Rising Edge *
Fifo Threshold	Empty
Output Drive	Enabled *

2.6. SAI3

Mode: Synchronous Slave

Mode: Synchronous Slave

2.6.1. Parameter Settings:

SAI A:

Synchronization Inputs	Synchronous with other SAI, SAI1 *
Protocol	Free
Audio Mode	Slave Receive
Frame Length (only Even Values)	64 *
Data Size	32 Bits *
Slot Size	DataSize
Output Mode	Stereo
Companding Mode	No companding mode
SAI SD Line Output Mode	Driven
First Bit	MSB First
Frame Synchro Active Level Length	32 *
Frame Synchro Definition	Channel Identification *
Frame Synchro Polarity	Active Low
Frame Synchro Offset	First Bit
First Bit Offset	0
Number of Slots (only Even Values)	2
Slot Active Final Value	0x00000003 *
Slot Active	User Setting *
Slot 0 Active	

Slot 1 Active	true *
Clock Strobing	true *
Fifo Threshold	Rising Edge *
Output Drive	Empty
SAI B:	Enabled *
Synchronization Inputs	Synchronous with other block of same SAI
Protocol	Free
Audio Mode	Slave Receive
Frame Length (only Even Values)	64 *
Data Size	32 Bits *
Slot Size	DataSize
Output Mode	Stereo
Companding Mode	No companding mode
SAI SD Line Output Mode	Driven
First Bit	MSB First
Frame Synchro Active Level Length	32 *
Frame Synchro Definition	Channel Identification *
Frame Synchro Polarity	Active Low
Frame Synchro Offset	First Bit
First Bit Offset	0
Number of Slots (only Even Values)	2
Slot Active Final Value	0x00000003 *
Slot Active	User Setting *
Slot 0 Active	true *
Slot 1 Active	true *
Real Audio Frequency	0
Error between Selected	0
Clock Strobing	Rising Edge *
Fifo Threshold	Empty
Output Drive	Enabled *

2.7. SPI1

Mode: Transmit Only Master

2.7.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
--------------	----------

Data Size	8 Bits *
First Bit	MSB First
Clock Parameters:	
Prescaler (for Baud Rate)	8 *
Baud Rate	24.0 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge
Advanced Parameters:	
CRC Calculation	Disabled
NSSP Mode	Enabled
NSS Signal Type	Software
Fifo Threshold	Fifo Threshold 01 Data
Tx Crc Initialization Pattern	All Zero Pattern
Rx Crc Initialization Pattern	All Zero Pattern
Nss Polarity	Nss Polarity Low
Master Ss Idleness	00 Cycle
Master Inter Data Idleness	00 Cycle
Master Receiver Auto Susp	Disable
Master Keep Io State	Master Keep Io State Enable *
IO Swap	Disabled

2.8. SPI2

Mode: Transmit Only Master

2.8.1. Parameter Settings:

Basic Parameters:	
Frame Format	Motorola
Data Size	16 Bits *
First Bit	MSB First
Clock Parameters:	
Prescaler (for Baud Rate)	8 *
Baud Rate	24.0 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge
Advanced Parameters:	
CRC Calculation	Disabled
NSSP Mode	Enabled
NSS Signal Type	Software

Fifo Threshold	Fifo Threshold 01 Data
Tx Crc Initialization Pattern	All Zero Pattern
Rx Crc Initialization Pattern	All Zero Pattern
Nss Polarity	Nss Polarity Low
Master Ss Idleness	00 Cycle
Master Inter Data Idleness	00 Cycle
Master Receiver Auto Susp	Disable
Master Keep Io State	Master Keep Io State Enable *
IO Swap	Disabled

2.9. SYS

Timebase Source: SysTick

2.10. TIM2

Clock Source : Internal Clock

2.10.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	99 *
Counter Mode	Up
Counter Period (AutoReload Register - 32 bits value)	999 *
Internal Clock Division (CKD)	Division by 2 *
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

2.11. TIM3

Clock Source : Internal Clock

Channel2: PWM Generation CH2

2.11.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	239 *
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	999 *

Internal Clock Division (CKD)	Division by 4 *
auto-reload preload	Disable
Trigger Output (TRGO) Parameters:	
Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)
Clear Input:	
Clear Input Source	Disable
PWM Generation Channel 2:	
Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

2.12. TIM8

Combined Channels: Encoder Mode

2.12.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 16 bits value)	0
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)
Trigger Event Selection TRGO2	Reset (UG bit from TIMx_EGR)

Encoder:

Encoder Mode	Encoder Mode TI1
____ Parameters for Channel 1 ____	
Polarity	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter	5 *
____ Parameters for Channel 2 ____	
Polarity	Rising Edge
IC Selection	Direct

Prescaler Division Ratio	No division
Input Filter	5 *

2.13. UART5

Mode: Asynchronous

2.13.1. Parameter Settings:

Basic Parameters:

Baud Rate	256000 *
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	FIFO mode disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

2.14. USART1

Mode: Asynchronous

2.14.1. Parameter Settings:

Basic Parameters:

Baud Rate	31250 *
Word Length	8 Bits (including Parity)

Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive Only *
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

*** User modified value**

3. System Configuration

3.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
ADC1	PA5	ADC1_INP19	Analog mode	No pull-up and no pull-down	n/a	
ADC2	PA4	ADC2_INP18	Analog mode	No pull-up and no pull-down	n/a	
RCC	PH0-OSC_IN (PH0)	RCC_OSC_IN	n/a	n/a	n/a	
	PH1-OSC_OUT (PH1)	RCC_OSC_OUT	n/a	n/a	n/a	
SAI1	PE2	SAI1_MCLK_A	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PE4	SAI1_FS_A	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PE5	SAI1_SCK_A	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PE6	SAI1_SD_A	Alternate Function Push Pull	No pull-up and no pull-down	Low	
SAI2	PD11	SAI2_SD_A	Alternate Function Push Pull	No pull-up and no pull-down	Low	
SAI3	PD9	SAI3_SD_B	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PD1	SAI3_SD_A	Alternate Function Push Pull	No pull-up and no pull-down	Low	
SPI1	PD7	SPI1_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	High *	
	PB3 (JTDO/TRACESWO)	SPI1_SCK	Alternate Function Push Pull	No pull-up and no pull-down	High *	
SPI2	PB10	SPI2_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Medium *	
	PB15	SPI2_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Medium *	
TIM3	PB5	TIM3_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	
TIM8	PC6	TIM8_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PC7	TIM8_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	
UART5	PB12	UART5_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB13	UART5_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
USART1	PB14	USART1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA10	USART1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
GPIO	PE3	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	KEY_IN0
	PC13	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	PAEN
	PC0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	High *	ADDR0
	PC1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	High *	ADDR1
	PC2_C	GPIO_Output	Output Push Pull	No pull-up and no pull-down	High *	ADDR2
	PA0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Medium *	BT_BUTT
	PA6	GPIO_Input	Input mode	Pull-up *	n/a	BT_LED
	PA7	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	KEY_IN1

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PB2	GPIO_Input	Input mode	Pull-up *	n/a	BT_bBUTT
	PE8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Medium *	BT_bLED
	PE12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Medium *	BT_PON
	PE14	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	MAX_LOAD
	PB11	GPIO_Input	Input mode	Pull-up *	n/a	SHIFT_B
	PD3	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	KEY_IN2
	PD5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TFT_DC
	PD6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Medium *	BSELECT
	PB4 (NJTRST)	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Medium *	TFT_RST
	PB6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TFT_CS
	PB7	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	KEY_IN3

3.2. DMA configuration

nothing configured in DMA service

3.3. BDMA configuration

nothing configured in DMA service

3.4. MDMA configuration

nothing configured in DMA service

3.5. NVIC configuration

3.5.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	15	0
TIM2 global interrupt	true	3	0
USART1 global interrupt	true	5	0
TIM8 update interrupt and TIM13 global interrupt	true	6	0
UART5 global interrupt	true	3	0
SAI1 global interrupt	true	0	0
SAI2 global interrupt	true	1	0
SAI3 global interrupt	true	2	0
PVD and AVD interrupts through EXTI line 16	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
ADC1 and ADC2 global interrupts	unused		
TIM3 global interrupt	unused		
SPI1 global interrupt	unused		
SPI2 global interrupt	unused		
TIM8 break interrupt and TIM12 global interrupt	unused		
TIM8 trigger and commutation interrupts and TIM14 global interrupt	unused		
TIM8 capture compare interrupt	unused		
FPU global interrupt	unused		
HSEM1 global interrupt	unused		

3.5.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	true	false
Debug monitor	false	true	false
Pendable request for system service	false	true	false
System tick timer	false	true	true
TIM2 global interrupt	false	true	true
USART1 global interrupt	false	true	true
TIM8 update interrupt and TIM13 global interrupt	false	true	true
UART5 global interrupt	false	true	true
SAI1 global interrupt	false	true	true
SAI2 global interrupt	false	true	true
SAI3 global interrupt	false	true	true

*** User modified value**

4. System Views

4.1. Category view

4.1.1. Current

Category view

Power Domain view



Choose filters ...

... by Power Domain

☐ D1

☐ D2

☐ D3

☒ None

Middleware									
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System Core	Analog	Timers	Connectivity	Multimedia	Security	Computing	Trace and Debug	Power and Thermal
BDMA	ADC1 ✓	TIM2 ✓	SPI1 ✓	SAI1 ✓				
CORTEX_M7 ✓	ADC2 ✓	TIM3 ✓	SPI2 ✓	SAI2 ✓				
DMA		TIM8 ✓	UART5 ✓	SAI3 ✓				
GPIO ✓			USART1 ✓					
MDMA								
IVIC ✓								
RCC ✓								
SYS ✓								

4.1.2. Without filters

Category view

Power Domain view

Choose filters ...

... by Power Domain

D1

D2

D3

None

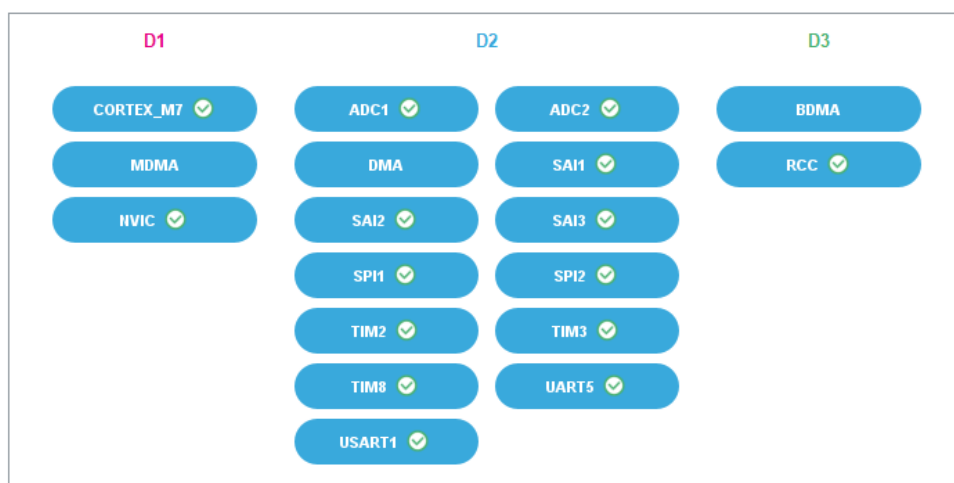
Middleware

System Core	Analog	Timers	Connectivity	Multimedia	Security	Computing	Trace and Debug	Power and Thermal
BDMA	ADC1 ✓	TIM2 ✓	SPI1 ✓	SAI1 ✓				
CORTEX_M7 ✓	ADC2 ✓	TIM3 ✓	SPI2 ✓	SAI2 ✓				
DMA		TIM8 ✓	UART5 ✓	SAI3 ✓				
GPIO ✓			USART1 ✓					
MDMA								
IVIC ✓								
RCC ✓								
SYS ✓								

4.2. Power Domain view

Category view

Power Domain view



5. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32h7_bsd1.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32h7_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32h7-svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers_stm32h7_series_product_overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval-tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32h7rs-lines-overview.pdf
Brochures	https://www.st.com/resource/en/brochure/brstm32h7.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32trust.pdf
Flyers	https://www.st.com/resource/en/flyer/flpowerstbd.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32h7rs.pdf
Security Bulletin	https://www.st.com/resource/en/security_bulletin/sb0023-eucleak-protection-statement-for-stmicroelectronics-certified-products-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an2606-stm32-

microcontroller-system-memory-boot-mode-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an3126-audio-and-waveform-generation-using-the-dac-in-stm32-products-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an3155-usart-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an3156-usb-dfu-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4286-spi-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4539-hrtim-cookbook-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4566-extending-the-dac-performance-of-stm32-microcontrollers-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4776-generalpurpose-timer-cookbook-for-stm32-microcontrollers-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4839-level-1-cache-on-stm32f7-series-and-stm32h7-series-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4891-stm32h72x-stm32h73x-and-singlecore-stm32h74x75x-system-architecture-and-performance-stmicroelectronics.pdf

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Application Notes https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5033-stm32cube-mcu-package-examples-for-stm32h7-series-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5073-receiving-spdif-audio-stream-with-the-stm32f4f7h7-series-stmicroelectronics.pdf

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Application Notes https://www.st.com/resource/en/application_note/an5354-getting-started-with-the-stm32h7-series-mcu-16bit-adc-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4760-quadspi-interface-on-stm32-microcontrollers-and-microprocessors-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4899-stm32-microcontroller-gpio-hardware-settings-and-lowpower-consumption-stmicroelectronics.pdf

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Application Notes https://www.st.com/resource/en/application_note/an5293-migration-guide-from-stm32f7-series-to-stmh74x75x-stm32h72x73x-and-stmh7a37bx-devices-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4991-how-to-wake-up-an-stm32-microcontroller-from-lowpower-mode-with-the-usart-or-the-lpuart-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4838-introduction-to-memory-protection-unit-management-on-stm32-mcus-stmicroelectronics.pdf

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pwm-shutdown-for-motor-control-and-digital-power-conversion-on-stm32-mcus-stmicroelectronics.pdf

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fdcan-peripherals-for-stm32-mcus-stmicroelectronics.pdf

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for related Tools
& Software

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Application Notes https://www.st.com/resource/en/application_note/an4891-stm32h72x-stm32h73x-and-singlecore-stm32h74x75x-system-architecture-and-performance-stmicroelectronics.pdf
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& Software

Application Notes https://www.st.com/resource/en/application_note/an5001-stm32cube-expansion-package-for-stm32h7-series-mdma-stmicroelectronics.pdf
for related Tools
& Software

Application Notes https://www.st.com/resource/en/application_note/an5014-stm32h7x3-smart-power-management-expansion-package-for-stm32cube-stmicroelectronics.pdf
for related Tools
& Software

Application Notes https://www.st.com/resource/en/application_note/an5033-stm32cube-mcu-package-examples-for-stm32h7-series-stmicroelectronics.pdf
for related Tools
& Software

Application Notes https://www.st.com/resource/en/application_note/an5056-integration-guide-for-the-xcubesbsfu-stm32cube-expansion-package-
for related Tools

& Software	stmicroelectronics.pdf
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an5360-getting-started-with-projects-based-on-the-stm32mp1-series-in-stm32cubeide-stmicroelectronics.pdf
& Software	stmicroelectronics.pdf
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an5361-getting-started-with-projects-based-on-dualcore-stm32h7-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf
& Software	stmicroelectronics.pdf
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an5394-getting-started-with-projects-based-on-the-stm32l5-series-in-stm32cubeide-stmicroelectronics.pdf
& Software	stmicroelectronics.pdf
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an5418-how-to-build-a-simple-usbp-d-sink-application-with-stm32cubemx-stmicroelectronics.pdf
& Software	
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an5426-migrating-graphics-middleware-projects-from-stm32cubemx-540-to-stm32cubemx-550-stmicroelectronics.pdf
& Software	
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an5564-getting-started-with-projects-based-on-dualcore-stm32wl-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf
& Software	
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an4865-lowpower-timer-lptim-applicative-use-cases-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
& Software	
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an5698-adapting-the-xcubestl-functional-safety-package-for-stm32-iec-61508-compliant-to-other-safety-standards-stmicroelectronics.pdf
& Software	
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an5731-stm32cubemx-and-stm32cubeide-threadsafe-solution-stmicroelectronics.pdf
& Software	
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an5450-stm32h7a37b3-lines-and-stm32h7b0-value-line-smart-power-management-expansion-package-for-stm32cube-stmicroelectronics.pdf
& Software	
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an4502-stm32-smbuspmbus-expansion-package-for-stm32cube-stmicroelectronics.pdf
& Software	

Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5952-how-to-use-cmake-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4635-how-to-optimize-lpuart-power-consumption-on-stm32-mcus-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5054-how-to-perform-secure-programming-using-stm32cubeprogrammer-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an6088-how-to-use-mce-for-encryptiondecryption-on-stm32-mcus-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an6179-how-to-integrate-the-stl-firmware-into-a-time-critical-user-application-stmicroelectronics.pdf
Design Notes & Tips	https://www.st.com/resource/en/design_tip/dt0117-microphone-array-beamforming-in-the-pcm-and-pdm-domain-stmicroelectronics.pdf
Errata Sheets	https://www.st.com/resource/en/errata_sheet/es0392-stm32h742xig-stm32h743xig-stm32h750xb-stm32h753xi-device-errata-stmicroelectronics.pdf
Datasheet	https://www.st.com/resource/en/datasheet/dm00387108.pdf
Programming Manuals	https://www.st.com/resource/en/programming_manual/pm0253-stm32f7-series-and-stm32h7-series-cortexm7-processor-programming-manual-stmicroelectronics.pdf
Reference Manuals	https://www.st.com/resource/en/reference_manual/rm0433-stm32h742-stm32h743753-and-stm32h750-value-line-advanced-armbased-32bit-mcus-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1163-description-of-wlcsp-for-microcontrollers-and-recommendations-for-its-use-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1204-tape-and-reel-shipping-media-for-stm32-microcontrollers-in-bga-packages-stmicroelectronics.pdf
Technical Notes	https://www.st.com/resource/en/technical_note/tn1205-tape-and-reel-

& Articles	shipping-media-for-stm8-and-stm32-microcontrollers-in-fpn-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1206-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-qfp-packages-stmicroelectronics.pdf
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Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1208-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-tssop-and-ssop-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1433-reference-device-marking-schematics-for-stm32-microcontrollers-and-microprocessors-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf
User Manuals	https://www.st.com/resource/en/user_manual/um2840-stm32h7-dualcore-series-safety-manual-stmicroelectronics.pdf
User Manuals	https://www.st.com/resource/en/user_manual/um2331-stm32h7-singlecore-series-safety-manual-stmicroelectronics.pdf
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